DIGITAL TWIN CHALLENGE 2026 · CEG, ANNA UNIVERSITY

# NERAM

## Thermal Digital Twin and Simulated Warming Controller

A patient-specific twin that forecasts one patient’s temperature signal 20–60 minutes ahead, with uncertainty.

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Proof-of-concept prototype, not a medical device. Simulated / assumed responses where marked.



THE PROBLEM

# During surgery, temperature drifts slowly — and a population curve says little about the patient on the table.

### One patient, own data

NERAM estimates the temperature-signal trajectory from this patient's own monitor data as the case goes on.

### Look ahead 20–60 min

A forecast with uncertainty bands, and a flag when the twin is unreliable (gaps, artefacts, start-up).

### Supervised warming

A simulated, clinician-supervised controller shows how a forecast could feed a warming workflow.

# What the prototype is, and is not

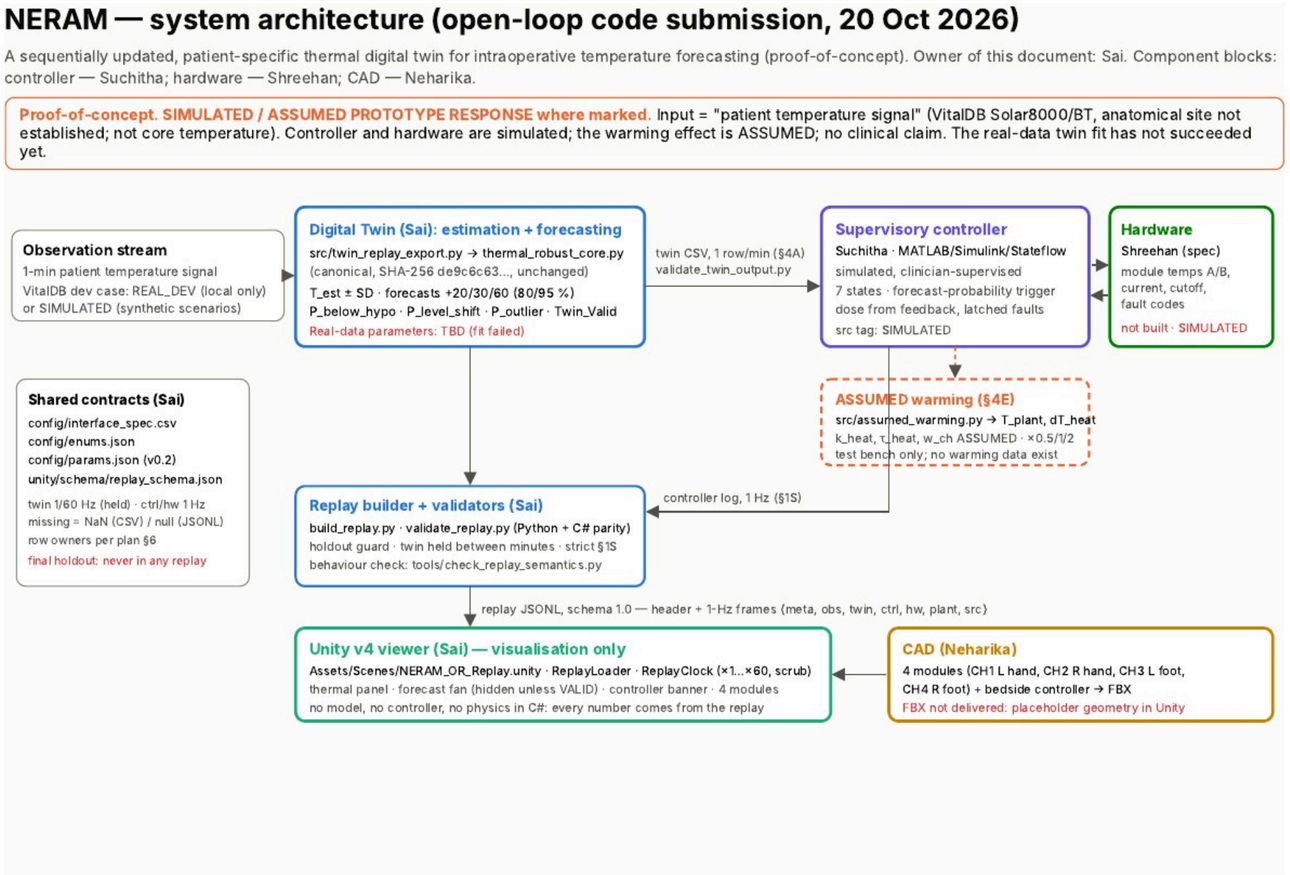
## It is

- A verified state-space twin implementation, exported per minute
- A reproducible, holdout-protected VitalDB development pipeline
- A simulated controller with hardware stand-ins, and a paper hardware design
- A Unity replay viewer (Windows app) that plays validated replays

## It is not

- A core-temperature measurement (the probe site is not established)
- A validated forecaster: the real-data fit has not succeeded yet
- A treatment device; no hardware has been built
- Clinically validated, certified, or evidence of any warming effect

Full text: docs/SAFETY_AND_LIMITATIONS.md




ARCHITECTURE

# Three parts, one replay contract

**Twin** = estimation and forecasting

**Controller** = simulated supervision

**Unity** = replay-only visualisation

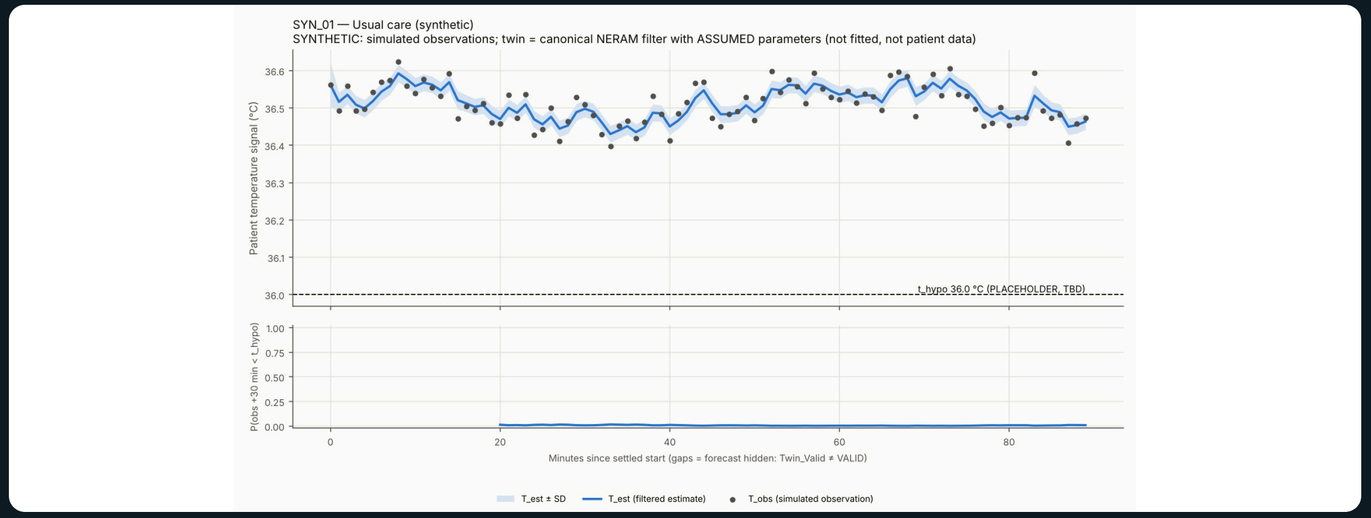
Joined by one JSONL replay contract (interface spec v1.0, frozen)

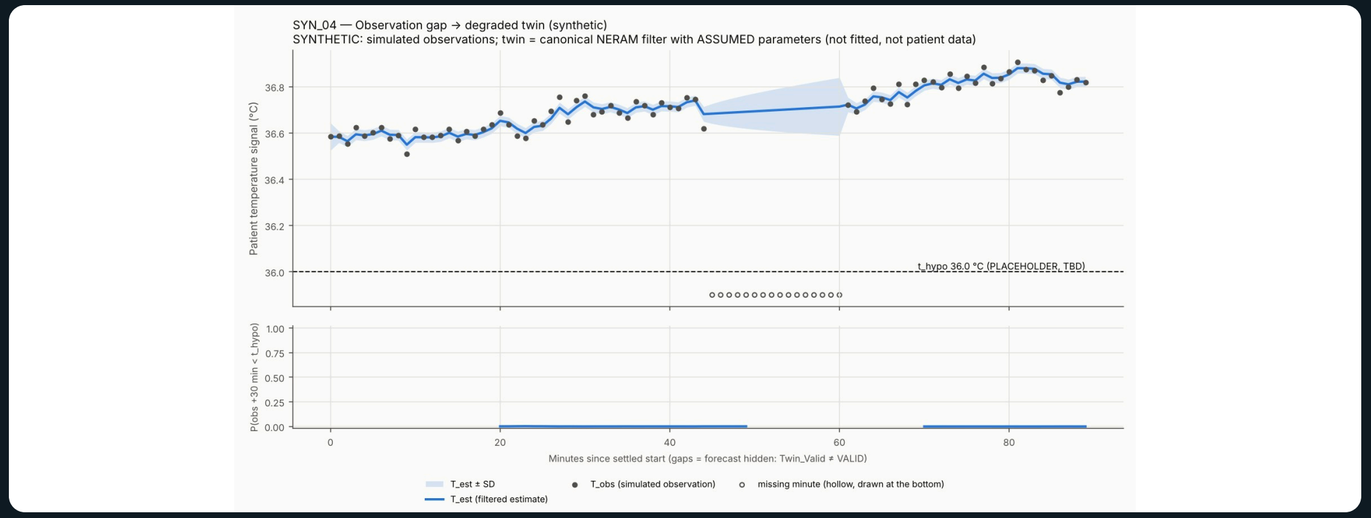
THE DIGITAL TWIN

# A filter that knows when not to trust itself

A switching state-space filter (GPB2) with jump and outlier mixtures estimates the signal each minute and forecasts +20 / +30 / +60 min with 80 % and 95 % bands.

Gaps and artefacts make the twin **DEGRADED**, and the forecast is hidden until it is valid again.



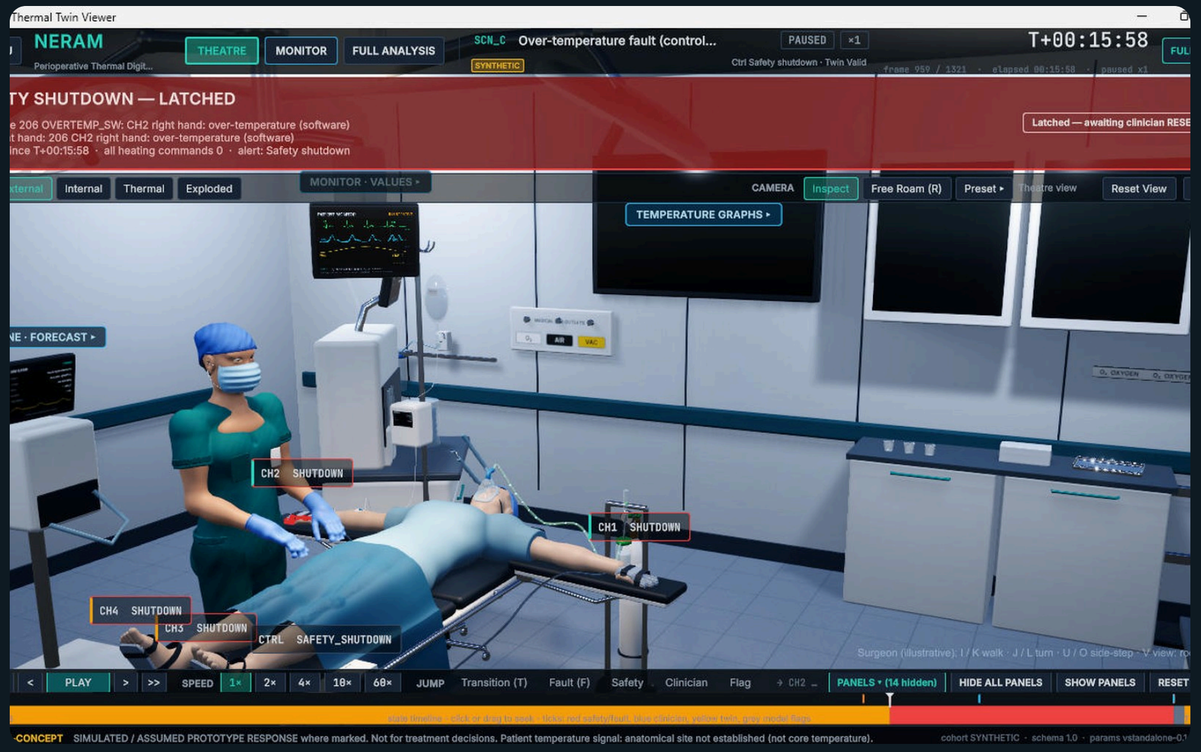
# Results, stated honestly

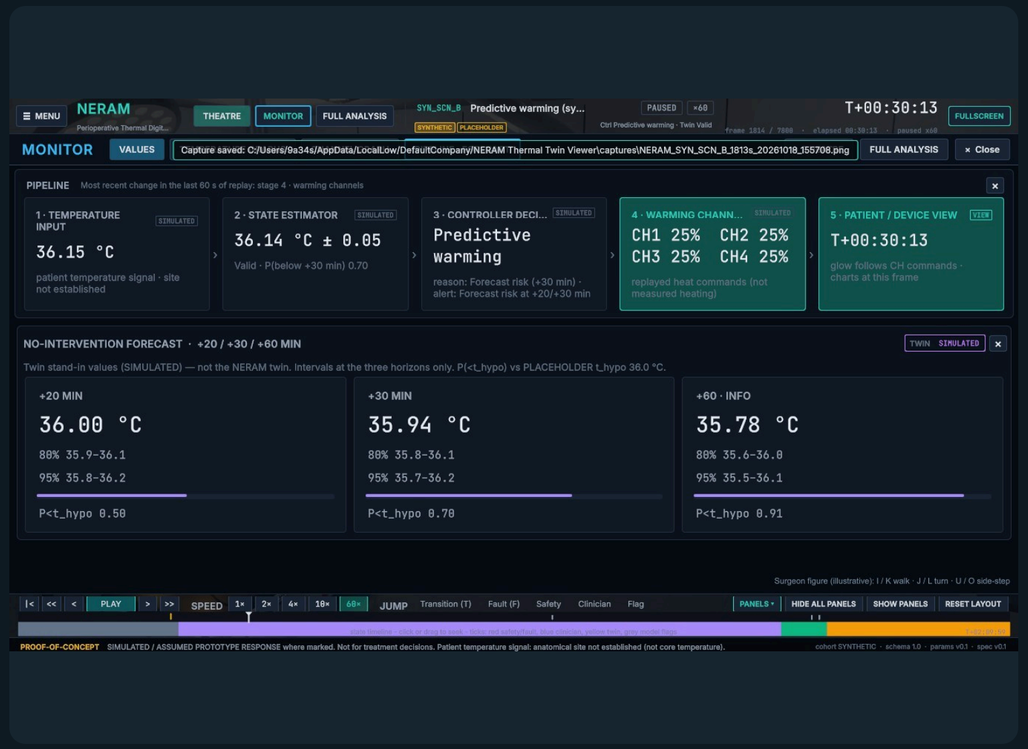
| Check | Status | What it shows |
|---|---|---|
| Synthetic verification | PASS | Closed-form limit, Monte Carlo P(below), no future leakage, Twin_Valid rules |
| Real-data development fit | FAILED | 0.1 °C quantised readings; 161 of 3,342 cases empty after the terminal cut |
| Correction | Not yet run | Written (C1–C4); needs review before it is run |
| Earlier exploratory test | Disclosed | Persistence beat the old model at +20/+30/+60 min (MAE 0.250 vs 0.352 °C at +20) |
| Final holdout | Frozen | 1,100 cases untouched; one-time evaluation pre-registered |

**No claim is made about real-data forecasting performance.**

UNITY REPLAY VIEWER · NERAM.EXE

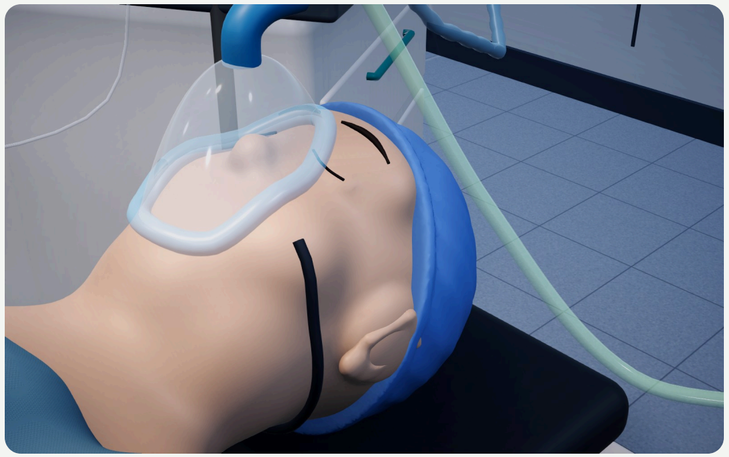
# See the twin, the controller and the devices together




5 replays from the controller tests (SCN_B–F, SIMULATED) + 11 SYNTHETIC UI replays · every value carries a source tag · Unity 6000.3, URP, UI Toolkit

# A theatre that reads at a glance



## Face mask and $O_2$ tubing

In the Internal view, oxygen particles flow through the airway into the lungs.



## $O_2$ cylinder at the bedside

Regulator, flowmeter and humidifier, tubing to the mask.



## Live monitor, walkable surgeon

Sweeping traces and the replay temperature; walk the surgeon with I J K L.

Illustrative scene props: no gas flow, breathing or physiology is modelled

SUPERVISORY CONTROLLER

# 7 states, 14 transitions, latched faults

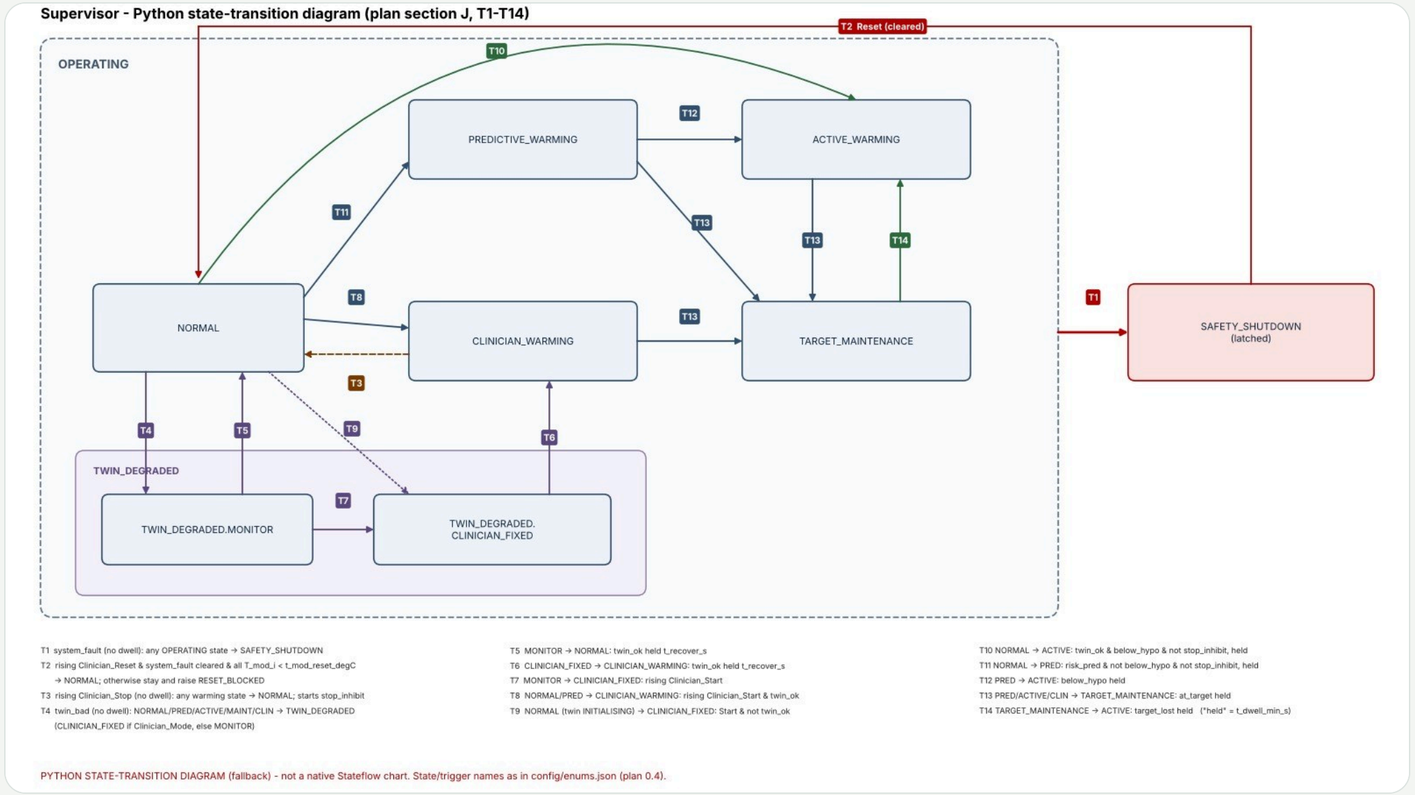
**18**

tests PASS (TC01–TC18)

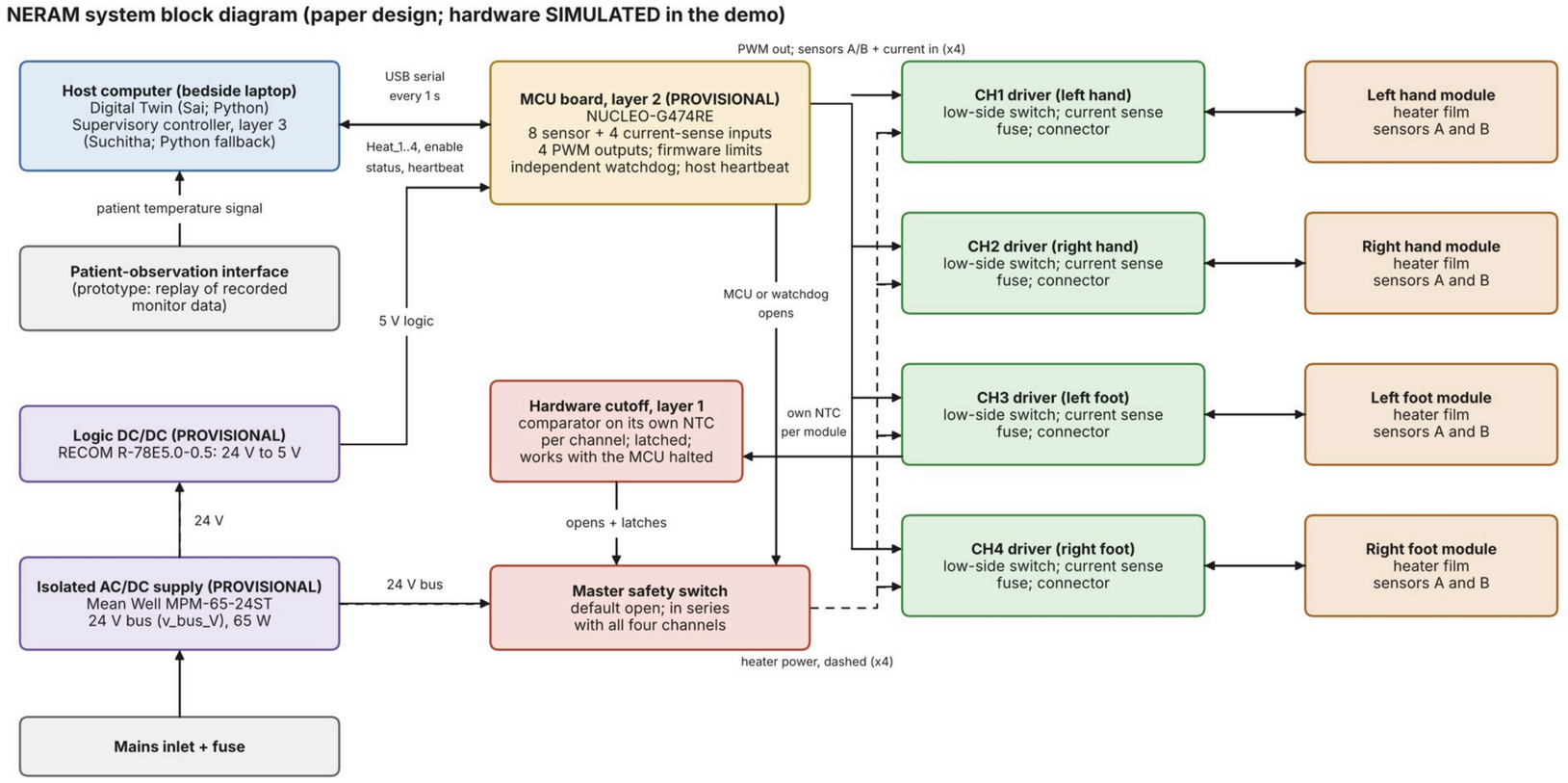
**1**

blocked (TC19, needs a development twin file)

Python fallback: MATLAB/Simulink/Stateflow were not available. Five blocks run every second.



# Hardware: a four-channel warming controller, on paper




Solid arrows: signals. Dashed arrows: heater power, which passes the master switch before reaching any channel.

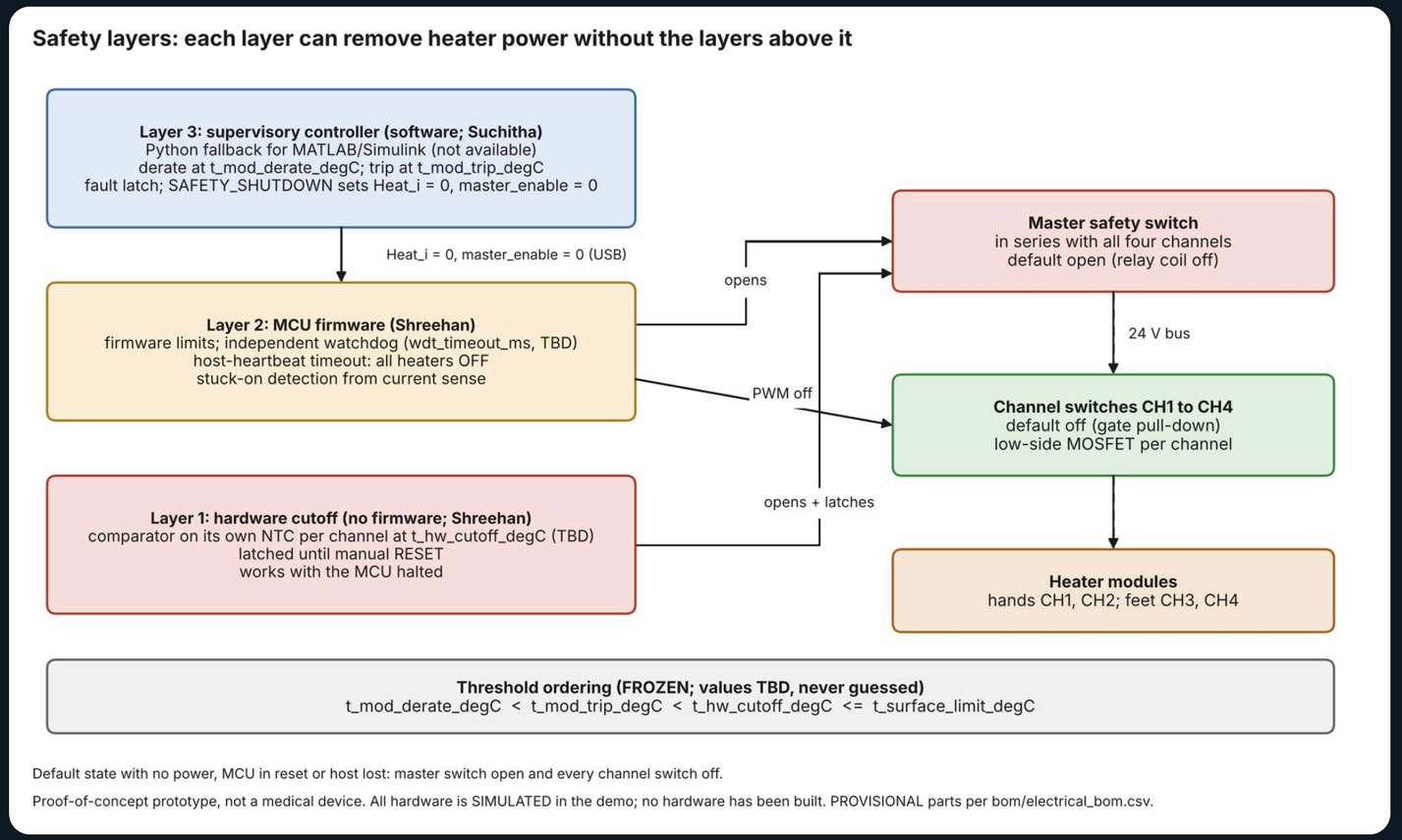
Proof-of-concept prototype, not a medical device. All hardware is SIMULATED in the demo; no hardware has been built. PROVISIONAL parts per bom/electrical_bom.csv.

SAFETY BY DESIGN

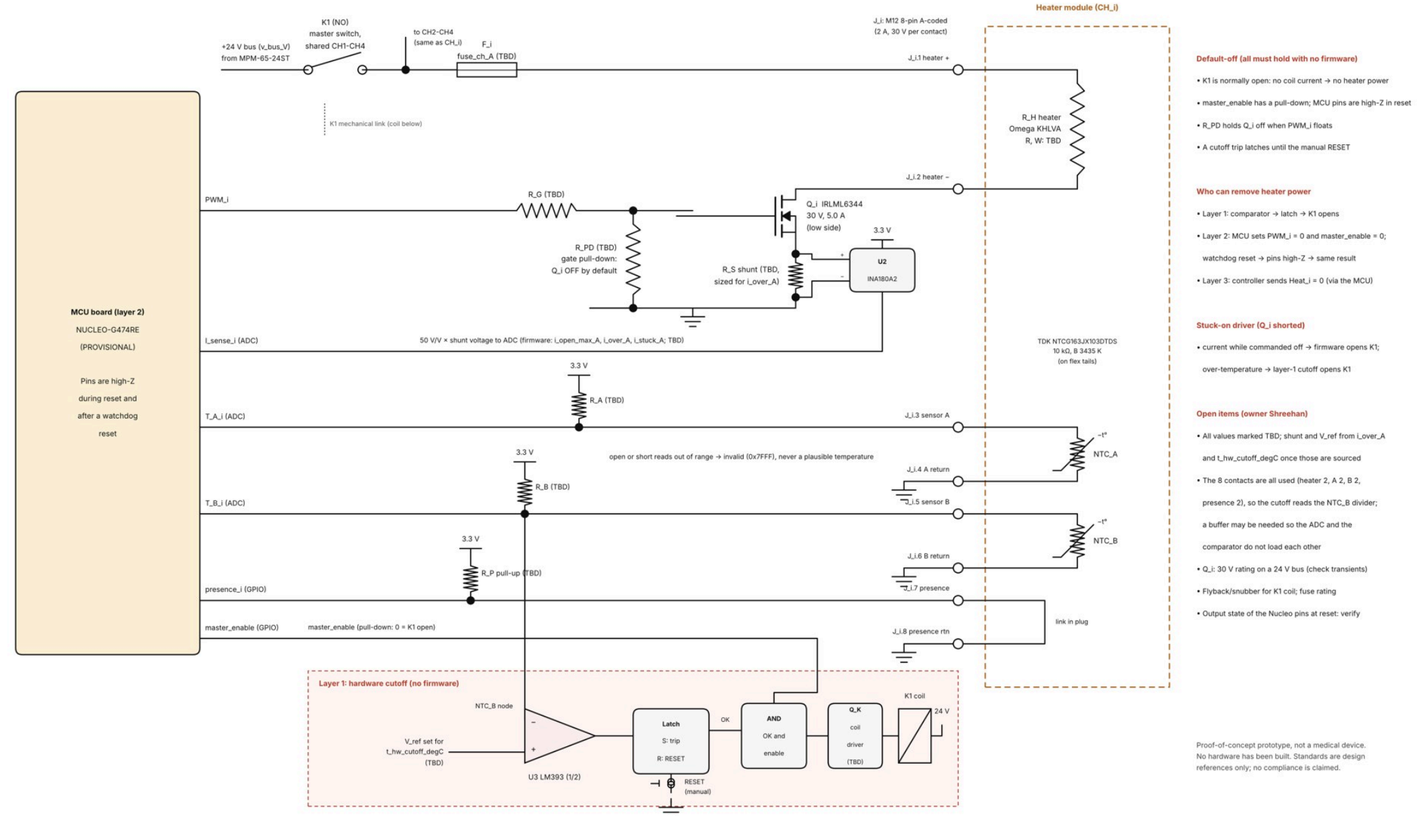
# Three layers, each able to cut heater power alone

1. Hardware cutoff: no firmware, latched until manual reset
2. MCU firmware: limits, watchdog, host heartbeat
3. Supervisory controller: derate, trip, shutdown

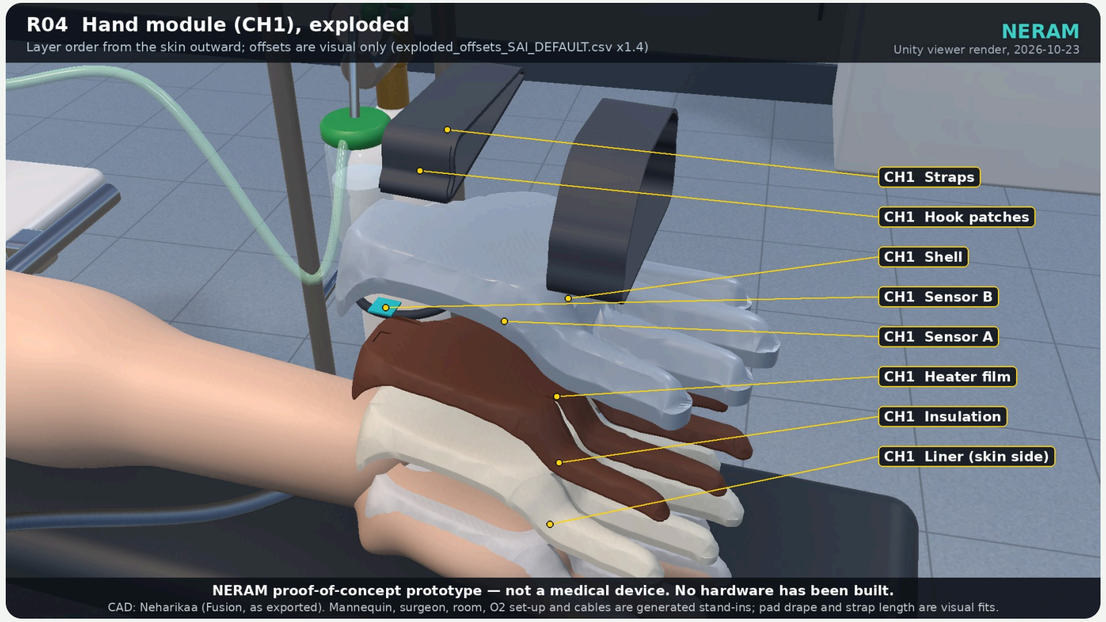
Threshold order is frozen; the values stay TBD until sourced, never guessed.



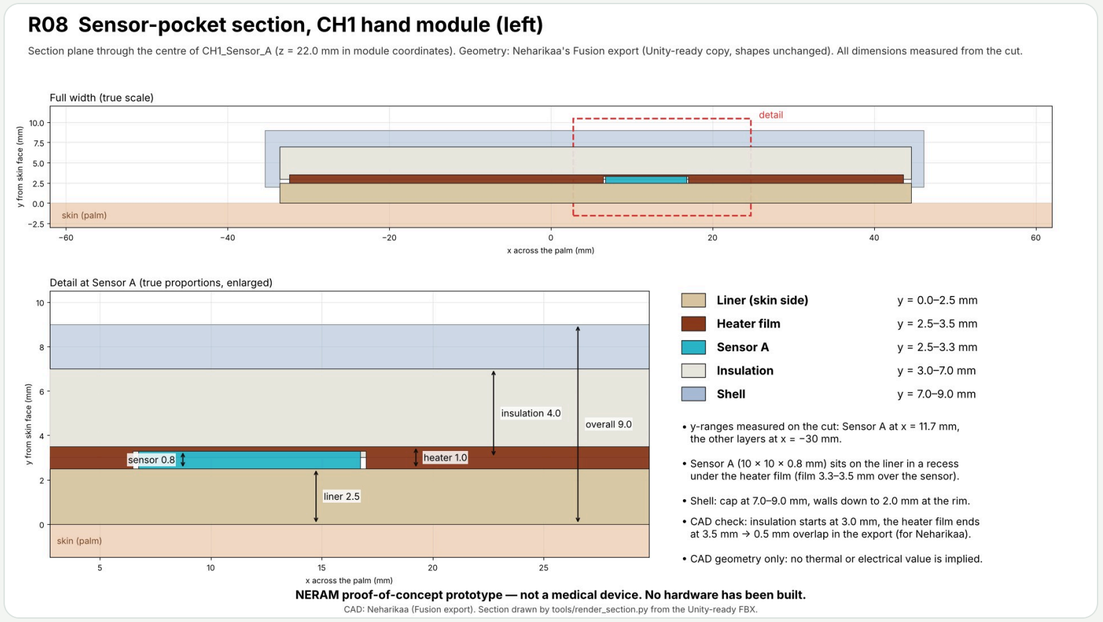
# One heater channel, default off



# CAD: layered hand and foot modules




Exploded CH1 module: liner, heater film, insulation, shell, two sensors, two straps




Section through Sensor A, measured from the CAD: 9 mm stack, sensor in a recess

# Open, tested, reproducible

**PASS**

Python test suite, on a fresh clone

**97 / 0**

Unity project checks pass / fail

**86 / 0**

Unity EditMode tests pass / fail

**0**

holdout files or case-level data published

Public GitHub repository · code Apache-2.0 · CAD, docs and figures CC BY 4.0 · VitalDB used under its data-use terms

Every script listed with its SHA-256 in docs/AUDIT_TRAIL.md · environment pinned · interface spec v1.0 frozen

# What comes next

### Make the twin real

Review and run the corrected development fit (rounding variance, empty cases), then the one-time holdout evaluation as pre-registered.

### Source the limits

Skin-surface limit and module thresholds from the governing standard and clinicians; heater power from the final coverage area.

### One channel on the bench

A supervised, current-limited single-channel build with no skin contact, checked against the simulated stand-ins.

A real device would also need standards testing, clinical evaluation and biocompatibility assessment.

# Thank you

**Saiyugan**
Lead · digital twin, integration, Unity viewer

**Suchita**
Supervisory controller, test bench

**Neharikaa**
CAD: warming modules, wearable, controller

**Shreehan**
Hardware, repository, licences, audit

CEG, Anna University · NERAM proof-of-concept, not a medical device